Part 1:

1.1. a)

65	65	65	65		
65	65	65	65		
65	65	65	65	65	

$$* \begin{bmatrix} -0.5 & -1 & -0.5 \\ -1 & 7.0 & -1 \\ -0.5 & -1 & -0.5 \end{bmatrix} =$$

65	65	..	..
..	..	..	..
.	.		

It does nothing.

b)

1	1	1	1	1	1
1	1	1	1	1	1
1	1	1	1	1	1
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

$$* \begin{bmatrix} -0.5 & -1 & -0.5 \\ -1 & 7.0 & -1 \\ -0.5 & -1 & -0.5 \end{bmatrix} =$$

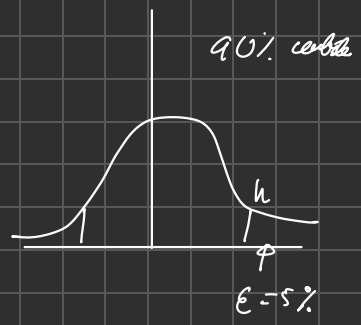
1	1	1	1
3	3	3	3
-2	-2	-2	-2
0	0	0	0

It amplifies edges.

$$g(x) < \varepsilon$$
$$g\left(\frac{h}{2}\right) \leq \varepsilon$$

1.2/

$$g(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \cdot \exp\left(\frac{-x^2}{2\sigma^2}\right)$$



$$g(h) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-h^2}{2\sigma^2}\right)$$

$$g\left(\frac{h}{2}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-\left(\frac{h}{2}\right)^2}{2\sigma^2}\right)$$

$$- \,||\, - \cdot \exp\left(-\frac{1}{4} \cdot \frac{h^2}{2\sigma^2}\right) \leq \varepsilon$$

$$\exp\left(\cdot \frac{\alpha^2}{2\sigma^2}\right) \leq \varepsilon \cdot \sqrt{2\pi\sigma^2} \quad / \ln$$

$$-\frac{\alpha^2}{2\sigma^2} \leq \ln\left(\varepsilon\sqrt{2\pi\sigma^2}\right)$$

$$\frac{\alpha^2}{2\sigma^2} \geq -\ln\left(\varepsilon\sqrt{2\pi\sigma^2}\right)$$

$$\frac{\alpha^2}{2\sigma^2} \geq \ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right)$$

$$\alpha \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right) \cdot 2\sigma^2}$$

$$\frac{h}{2} \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right) \cdot 2\sigma^2}$$

b) $$\alpha \geq \sqrt{\ln\left(\frac{1}{\varepsilon\sqrt{2\pi\sigma^2}}\right)\cdot 2\sigma^2}$$

$$\alpha \geq \sqrt{\ln\left(\frac{1}{\frac{1}{256}\sqrt{2\pi\, 3^2}}\right)\cdot 2\cdot 3^2}$$

$$\alpha \geq 8$$

$$\underline{K = 2k+1 = 17}$$